Water Surface Instruments DIY and Open Hardware for Plastic Pollution Tracking

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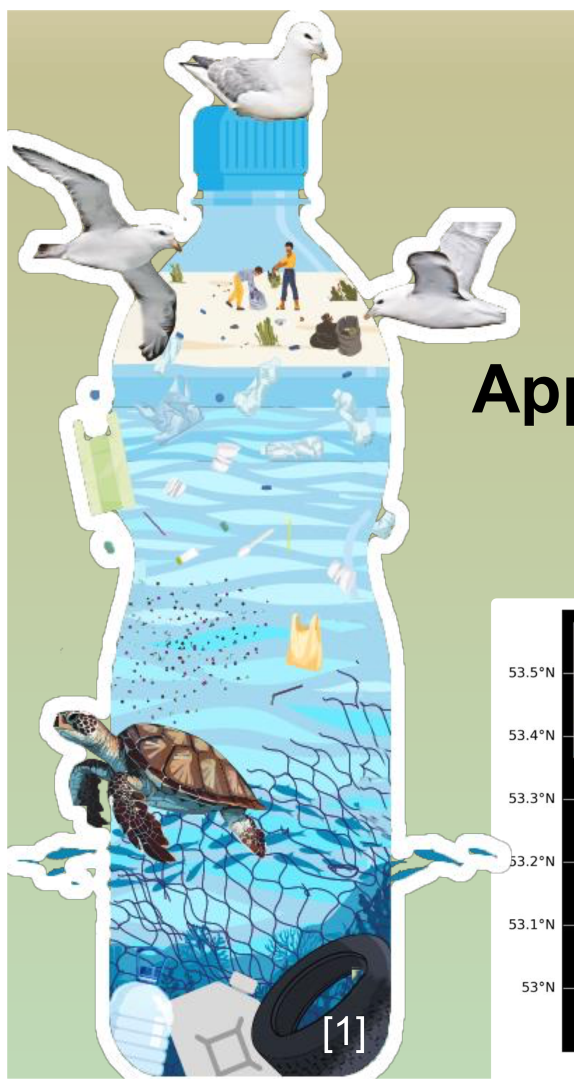
Floating Marine Macroplastics Research

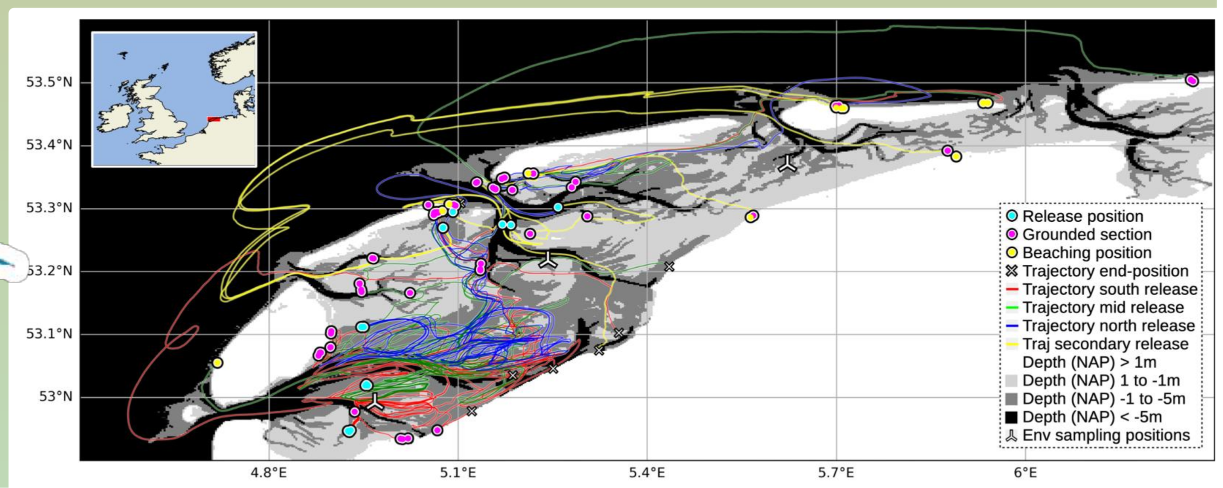


How does it enter the ocean?
How is it transported in the water?
Where does it accumulate?

Approaches

Dispersal simulations
Drifter campaigns

PlasticParcels



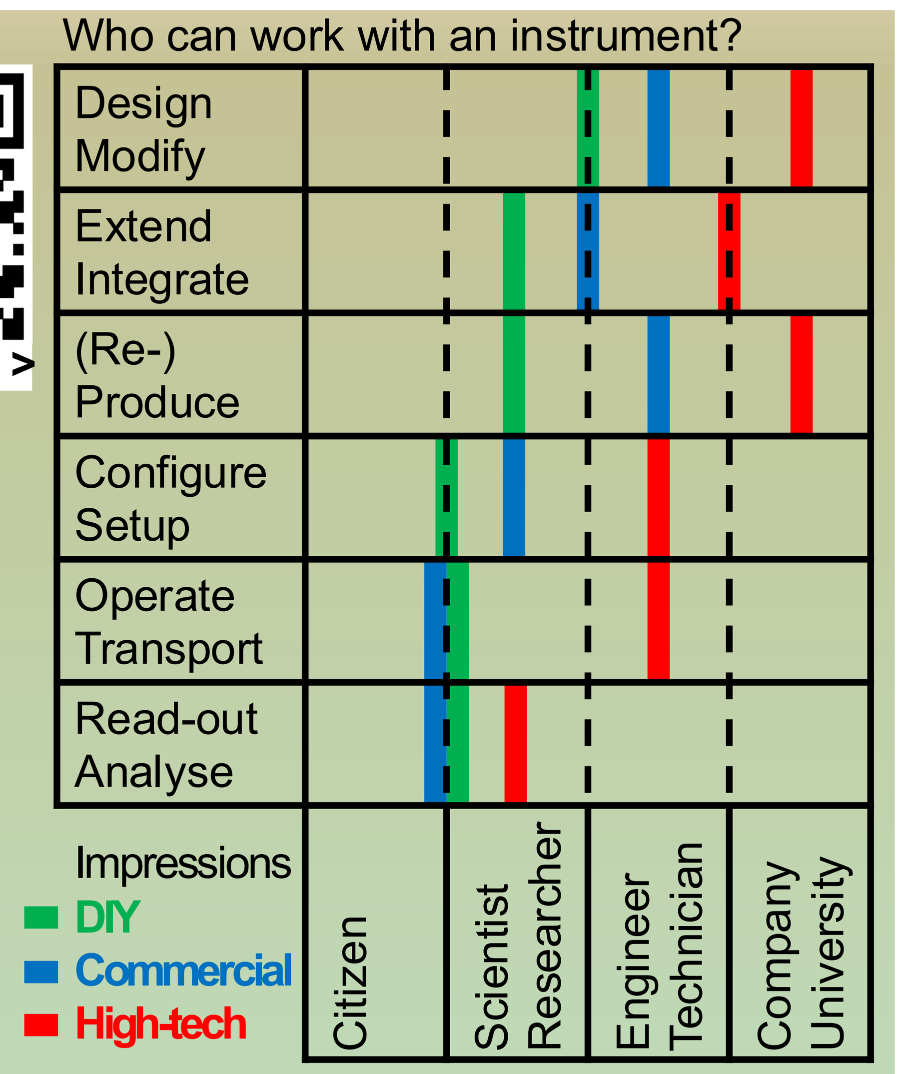
DIY and Open Science / Hardware

Principles (FAIR)
- Findable
- Accessible
- Interoperable
- Reusable

Goals
- Reproducibility
- Collaboration
- Equitable participation

Challenges
- Managing complexity
- Liability and Safety
- Secondary applications

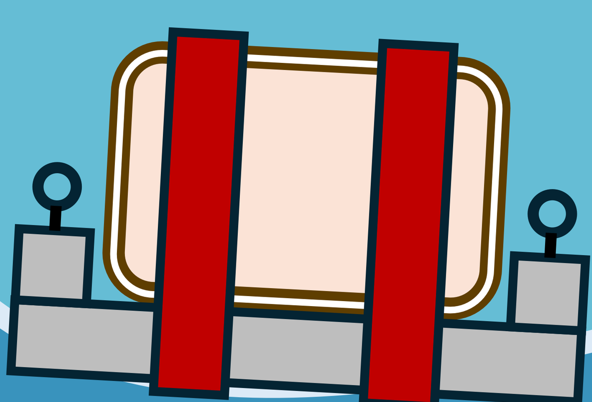
Poll for table >

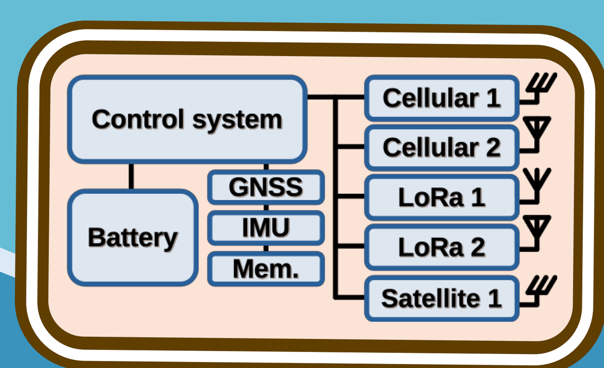


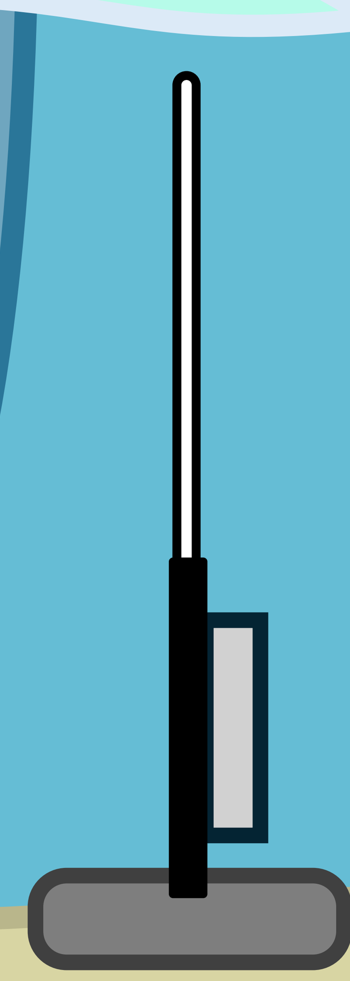
Mass Producible DIY Surface Drifter

Hull	Dynamics	15cm 700g	15cm 700g	20cm 700g	20cm 2000g
Hull	Waterproof enclosure	Wooden plate	Bottle	Pipe PVC, (PMMA)	Mold
Electronics	Sensing	-	GNSS	IMU & Compass	Light, Temperature
Electronics	Telecom-munication	-	LoRa	Cellular	Satellite Kinéis Globalstar Iridium (Lacuna)
Electronics	Power	-	Battery Alkaline, NiMH	Solar	
Electronics	Local storage	-	Microcontroller	SD card	Memory IC

Criteria | large number productions | simple open hardware technology | mimic floating macroplastic | low cost | environmentally responsible and safe | modules or custom PCB integrated |

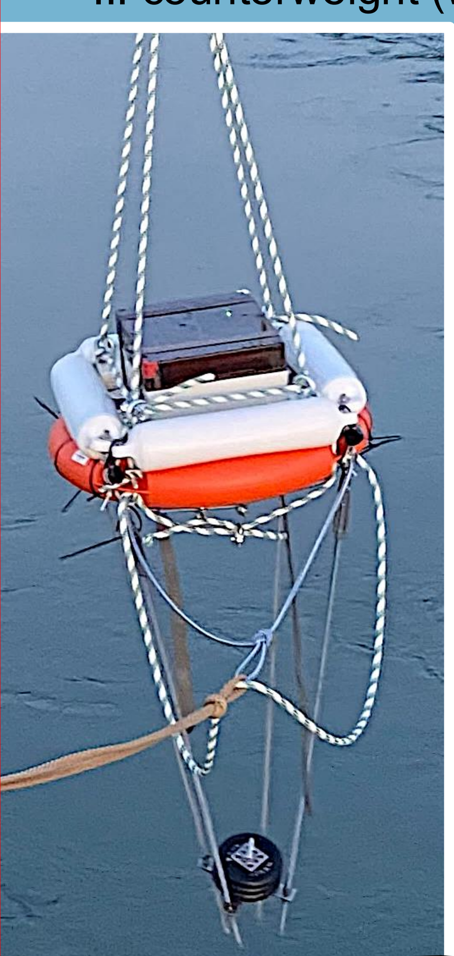






Buoy

Platform of 30x24cm
Portable (2 stages assembly)
Adjustable dynamics with ...
... float (fenders)
... counterweight (weights, position)



> Dimensions
1m high, 0.6m wide
Weight 11.5kg + 4kg

> Materials
Safety buoy
Fenders
Disc weights
Threaded rods
Turnbuckles & eyes
Rope

> Production
Mech. assembly
(Basic steelwork)

Carrierplate & Case

Buoyant testbed-carrierplate combination
Carrierplate to position ...
... on water surface by tolling
... in-air close to ship deck
Custom mounting system for case interior

> Dimensions
Case 30x24x12 cm
Plate 38x32x10 cm
Weight 2-5kg

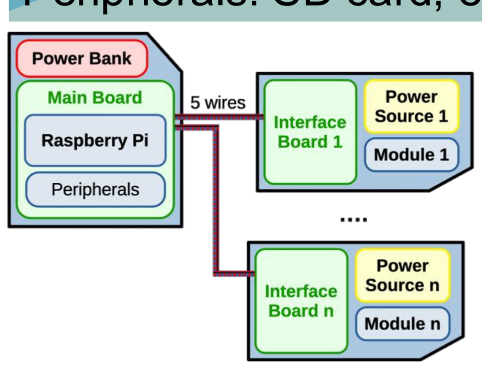
> Materials
Betonplex plate
Eyes & Bolts
Rod clamps
Ratchet straps
Case IP68
Acrylic sheet(s)

> Production
Woodwork & painting
Mech. Assembly
Laser-cutting

Control & Logging System

Main system operates multiple test modules
Interface per module
- Power-control
- 2-wire Serial (I2C)

Peripherals: SD card, env. sensors, I/O



> Dimensions
Main board ~11cm
Module brd. ~6cm

> Materials
Raspberry Pi
Modules (Arduino)
Proto-/Bread-board
Powerbank
Sensors & SD card
(Wireless charger)

> Production
Hand-soldering
Python coding

Receiver Station

Antenna + gateway with mains power supply
Weatherproof (in particular wind)
Temporary installation (~1 month)

> Dimensions
1m high, 0.5m wide
Weight 20kg

> Materials
Parasol stand
Metal tube
Hose clamp (Wing)
Sand / Soil
Gateway & Antenna
Power cable & Case

> Production
Basic metalwork
Server (Chirpstack)

[1] Figure modified, OSPAR, https://www.ospar.org/
Designed with elements from SVG Silh, https://svgsilh.com/